

Worked Answer Key

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1. $3/8$: 3 of 8 equal parts are shaded. 2. Numerator 5 counts the parts taken; denominator 7 tells how many equal parts make the whole. 3. Any bar split into 6 equal parts with 4 shaded; $4/6$. Yes, the parts must be equal in size so that each represents $1/6$ of the whole.

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1. $3/5$ of 20 = $20 \div 5 \times 3 = 12$. Draw 5 groups of 4 counters and circle 3 groups: 12 counters.
2. Blue $7/12$; not blue $5/12$. 3. $24 \div 4 = 6$ students.

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1. $1/9$. 2. Divide a strip into 8 equal pieces. Each is $1/8$ of the strip, and all 8 pieces together make 1 whole. 3. For the same whole, $1/3 = 2/6 > 1/6$.

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1. $7/4 = 1\frac{3}{4}$ (4 goes into 7 once, remainder 3). 2. $2\frac{3}{5} = (2 \times 5 + 3)/5 = 13/5$. 3. One whole plus 1 of 2 parts; $1\frac{1}{2} = 3/2$.

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1. Six equal intervals from 0 to 1; the fourth tick after 0 is $4/6$. 2. $3/8$. 3. $5/4 = 1\frac{1}{4}$; mark one whole and one fourth.

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1. Multiply by 4: $\frac{2}{3} = \frac{8}{12}$. 2. Multiply by 3: $\frac{5}{8} = \frac{15}{24}$. 3. Examples: $\frac{6}{14}$ and $\frac{9}{21}$.

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1. Divide by 6: $\frac{12}{18} = \frac{2}{3}$. 2. Divide by 5: $\frac{20}{25} = \frac{4}{5}$. 3. Yes: $18 \div 6 = 3$ and $24 \div 6 = 4$, so $\frac{18}{24} = \frac{3}{4}$.

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1. Split each half of the same whole into 4 equal pieces. The whole now has 8 equal pieces, and one half covers 4 of them: $\frac{1}{2} = \frac{4}{8}$. 2. Examples: $\frac{1}{2}$ and $\frac{6}{12}$ are equivalent to $\frac{3}{6}$; other equivalent fractions are valid. 3. True: multiply both numerator and denominator by 2: $\frac{2 \times 2}{5 \times 2} = \frac{4}{10}$.

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1. Multiply by 5: $\frac{3}{4} = \frac{15}{20}$. 2. Multiply by 3: $\frac{7}{9} = \frac{21}{27}$. 3. Multiply by 3: $\frac{4}{11} = \frac{12}{33}$.

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1. Divide by 2: $\frac{6}{10} = \frac{3}{5}$. 2. Divide by 4: $\frac{8}{12} = \frac{2}{3}$. 3. Divide by 5: $\frac{15}{35} = \frac{3}{7}$.

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1. Divide by 4: $\frac{16}{20} = \frac{4}{5}$. 2. Divide by 7: $\frac{21}{28} = \frac{3}{4}$. 3. Yes; 9 and 13 have no common factor greater than 1.

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1. Yes: they ate fractions of the same pizza, and dividing both numerator and denominator of $\frac{2}{6}$ by 2 gives $\frac{1}{3}$. 2. Six eighths of a cup: $\frac{3}{4} = \frac{6}{8}$ cup. 3. Divide by 2: $\frac{6}{10} = \frac{3}{5}$ mile.

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1. $\frac{2}{3} = \frac{8}{12}$, so $=$. 2. Multiply both numerator and denominator by 3: $\frac{5}{6} = \frac{15}{18}$, so $=$. 3. Neither is greater: multiplying both numerator and denominator of $\frac{3}{5}$ by 3 gives $\frac{9}{15}$, so they are equal.

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1. Draw two equal-size rectangles. Split one into 3 equal parts with 2 shaded; split the other into 9 equal parts with 6 shaded. Both show the same amount: $\frac{2}{3} = \frac{6}{9}$. 2. Draw a rectangle divided into 4 equal parts and shade 3. Split each fourth into 3 equal smaller parts. Label the shaded amount $\frac{3}{4}$ and $\frac{9}{12}$: the same area is shaded. 3. Draw 15 objects in 5 groups of 3. Shade all the objects in 2 groups: 2 of 5 groups and 6 of 15 objects are shaded, so $\frac{2}{5} = \frac{6}{15}$.

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1. Split every part of the same whole into 3 equal smaller parts. There are now 3 times as many parts in the whole and 3 times as many selected parts, but the selected amount is unchanged. 2. The denominator also changes; $\frac{4}{8}$ and $\frac{1}{2}$ are equal, not greater. 3. Multiply both numerator and denominator by 4: $\frac{5}{6} = \frac{20}{24}$.

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1. $9/4 = 2\frac{1}{4}$; converting back gives $(2 \times 4 + 1)/4 = 9/4$. 2. $1\frac{2}{3} = (1 \times 3 + 2)/3 = 5/3$.
3. Yes: $2\frac{1}{2} = (2 \times 2 + 1)/2 = 5/2$.

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1. If 4 counters are $2/3$, one third is 2 counters, so the whole is $2 \times 3 = 6$ counters. 2. $6 \div 3 = 2$ per fifth; $2 \times 5 = 10$ tiles. 3. $15 \div 5 = 3$ per eighth; $3 \times 8 = 24$ cm.

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1. Valid in this case: $3 + 3 = 3 \times 2$ and $5 + 5 = 5 \times 2$, so Jordan has doubled both numerator and denominator. Thus $3/5 = 6/10$. Adding arbitrary numbers does not generally make equivalent fractions. 2. Not simplest: $6/8$ divides by 2 to $3/4$. 3. The denominator was multiplied by 2 without multiplying the numerator; correct form is $2/7 = 4/14$.

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1. $24/32$ divides by 8 to $3/4$; denominator 20 requires multiply by 5: $15/20$. 2. $18/30$ divides by 6 to $3/5$; denominator 5 fraction is $3/5$. 3. $3/4 < 4/4 < 3/2$ (0.75, 1, 1.5).

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1. $18/24$ divides by 6 to $3/4$; multiply by 2 for denominator 8: $6/8$. 2. Examples: $8/10$ ($\times 2$), $12/15$ ($\times 3$), $20/25$ ($\times 5$); each multiplier applies to both numerator and denominator. Other sets of three different equivalent fractions are valid. 3. Multiply numerator and denominator by the same nonzero whole number; $\frac{7 \times 4}{10 \times 4} = \frac{28}{40}$, so the missing numerator is 28.